

ChimeraX Command Line Reference Sheet

Jason Porta

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Table of Contents:

1. Learning ChimeraX
 - 1.1. ChimeraX User Guide
 - 1.2. Tutorials
 - 1.3. Ohi Lab Computation Guide
 - 1.4. Mailing lists
 - 1.5. Miscellaneous
2. General Features
 - 2.1. Basic Viewing Functions
 - 2.2. Molecule Display
 - 2.3. Nucleotides
 - 2.4. Graphics
 - 2.5. Map Display Features
 - 2.6. Markers
 - 2.7. Right Mouse
 - 2.8. Tools

General commands:

#Help pages:

```
$ help
$ help [command]
```

#Save files:

```
$ save scaffold.pdb model-spec
$ save hybrid_1.mrc model-spec
$ save chimera.cxs
$ save coordinates.pdb model-spec displayedOnly true
  selectedOnly True
```

#Fetch files from various online databases:

```
$ open 4893 fromDatabase emdb
$ open 6ALH fromDatabase pdb
```

1 Atom and object selection syntax:

#Select residues in model 1, chains I and J:

```
$ select #1/i:830-945,1037-1058/j:24-112
$ select (add | subtract | intersect) spec
$ select @H      --select all hydrogens
```

#Save selections:

```
$ name frozen mcore sel
$ name delete mcore      --forgets the selection
```

#Deselect all:

```
$ ~select
```

#Select residues of all currently selected atoms:

```
$ select sel residues true
```

#Show only bonds from salt bridges:

```
$ hbonds sel reveal true showDist true saltOnly true
```

3 Volume operations:

#Extract map density around atoms:

```
$ color zone #5 near #4 distance 5
$ volume splitbyzone #5
```

```

#Volume resampling:
    $ volume resample #1 onGrid #4
    $ volume zone #6 nearAtoms #5 range 5

#Change volume step and level (all maps):
    $ volume step 1
    $ volume level 0.1

#Hide disconnected density:
    $ surface dust #2 size 5

#Scale maps and calculate difference:
    $ volume subtract #1 #3 minRMS True onGrid #1

#Transparency: make (a)toms and (c)artoons 80% transparent:
    $ transparency #1 80 target ac

#Depth cue:
    $ lighting depthCue true depthCueStart 0.4 depthCueEnd 0.7

#Superpose two structures globally:
    $ mm #2 to #1 showAlignment True

#Morph between two structures:
    $ morph #1,2 wrap true frames 60

#Draw an icosahedron:
    $ shape icos divisions 1 mesh True radius 400

#Hide cartoon rendering of chains g-k in model #1:
    $ hide #1/g-k cartoons

#Align by LSQ:
    $ align #2 toAtoms #1 matchChainIds True reportMatrix True

#Assign secondary structure (0=coil, 1=alpha helix, 2=beta strand):
    $ setattr #3/j:365-369 res ss_type 2

```

- **Movies:**

#Open multiple maps as a volume series and play the series:

```
$ open *.mrc vseries true
```

#The colorRange option is for coloring the map with 'color zone':

```
$ vseries play #1 direction oscillate normalize true loop true  
cacheFrames 60 colorRange 20
```

#Rock object back and forth:

```
$ turn y 25 300 rock 150
```

#Record movie with 4 commands on one line separated by semicolons:

```
$ movie record; turn y 25 300 rock 150; wait 300; movie encode  
~/Desktop/movie2.mp4
```

#Record a movie at high resolution with no motion:

```
$ movie record size 4096,4096; wait 5; movie encode  
~/Desktop/gbp.mp4 framerate 1
```

#Rotate a model clockwise about a defined point along the x-axis:

```
$ turn x -7 center 178.38,157.92,156.14 coordinateSystem 7  
models 9
```

#Apply model position to another model:

```
$ view position model-spec sameAsModels ref-model-spec
```

#Print matrix of current camera and model position:

```
$ view matrix
```

#Apply transformation matrix:

```
$ view matrix mod 1,1,0,0,0,0,1,0,0,0,0,1,0
```

#CAV1 8S movie:

```
$ movie record; turn x 1 90; wait 180; turn y 1 60; wait 180;  
turn x 1 90; wait 180; transparency #2 80; wait 180; turn x 1  
180; wait 180; movie encode quality highest output  
Cav1_8S-movie.mp4; movie stop
```

5. Coloring operations:

#Apply colors from model to density map:

```
$ color radial #1 center 0,0,0
$ color sel yellow
```

#Sequentially color the atoms and cartoon with the following palette:

```
$ color sequential #1 chains target ac palette
6633FF:660099:CC66FF:9966FF:3300FF:6666FF:6699FF:0066FF:00
CCFF:99CCFF:333399cre
```

#CAV1 8S figure coloring scheme:

```
$ color #1/a-k:49-60 #ffff33; color #1/a-k:61-67 #3399ff;
color #1/a-k:68-75 #ff3333; color #1/a-k:76-80 #3399ff;
color #1/a-k:81-101 #00e673; color #1/a-k:102-134 #9933ff;
color #1/a-k:135-169 #999999; color #1/a-k:170-end #33ffff
```

6. Command files:

#Open and run a ChimeraX command file (*.cxc):

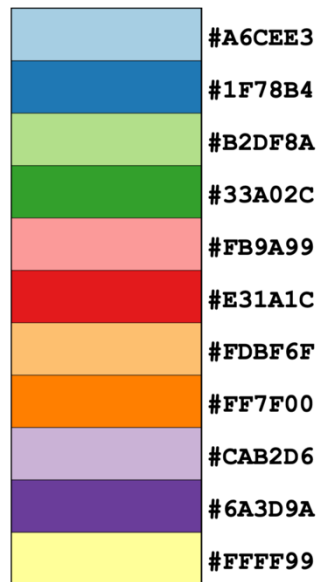
```
$ open initialize_commands.cxc
```

#Example of a ChimeraX command file:

```
#Initialize model and maps
#BEGIN
select #3; volume sel style surface
~select
hide #1 target s
wait 30
color #1/b-k #3399FF; color #1/a #FF3399
wait 30
show #!1 target ac; color #1 byhetero;
wait 30
hide #1 models
wait 30
show #2; show #3
wait 30
transparency #2 0; transparency #3 50
wait 30
view orient
#END
```

5. ChimeraX color palettes from colorbrewer.org:

- 11-class paired: ([colorbrewer 11-class paired](#)):



- 11-class RdYlBu ([colorbrewer 11-class RdYl1Bu](#)):
 - color blind friendly

